

COL1000: Lab 4

Semester 2, 2025–26

Instructions Upload your code to Gradescope as: `<entryNumber>_q<quesNumber>.py`. After uploading, run the autograder to check that all tests pass. Make sure not to include any custom prompt text in `input()` calls, and any extra text in `print()` statements. You must not use list/array conversion, or built-in operations such as `split`, `splitlines`, etc.

Question 1: Entry number Write a program that reads a student ID number and converts it to entry number. The format is: "20yy" + first 3 chars (not in small) + last 4 digits.

For example, if the input is `am1246789`, then the output must be `2024AM16789`.

Question 2: Mini Sudoku (4×4) Write a program that reads a 4×4 Sudoku grid and outputs `True` if the grid is valid, and `False` otherwise. A grid is considered *valid* if every row, every column, and each 2×2 block contains the numbers 1 through 4 exactly once.

For example, your code should output `True` for the following input.

```
14|32
23|41
--|--
32|14
41|23
```

Question 3: Last Longest Word Write a program that reads a sentence and outputs the last word that has the maximum length (words separated by spaces). For example:

```
Input: "My favourite subjects are philosophy arts literature and maths"
Output: "literature"
```

Question 4: Longest Palindrome Write a program that reads a single word (all lowercase letters) and prints the length of the longest palindromic substring. Single characters are valid palindromes of length 1.

For example, if input is `mississippi`, then the output must be `7` (corresponding to substring `ississi`).